Persian version: ../03-roadmap-and-future-work.html

Meeting Assistant
Roadmap & Future Work

Document 3 of 3 · Product roadmap, advanced capabilities, and research directions · Project overview · Implementation · CHANGELOG

This document gathers everything that comes *after* the MVP: the product roadmap by sprint, three advanced differentiating capabilities (facilitator guide, document alignment, sentiment analysis), the operational quality cycle (MLOps), and a set of longer-horizon research ideas. It complements the "future ideas" in Document 1 (vision) and Document 2 (current system).

> **Status guide.** Sprint 1 (the MVP) is complete. Among advanced capabilities, **facilitator guide** is already implemented (`GET /api/meetings/{id}/facilitation`); document alignment and sentiment analysis are designed but not yet built.

> **Permanently out of scope:** Live streaming transcription during meetings and professional speaker diarization. Single-pass Gemini transcription from uploaded/recorded audio covers product needs without a separate STT pipeline.

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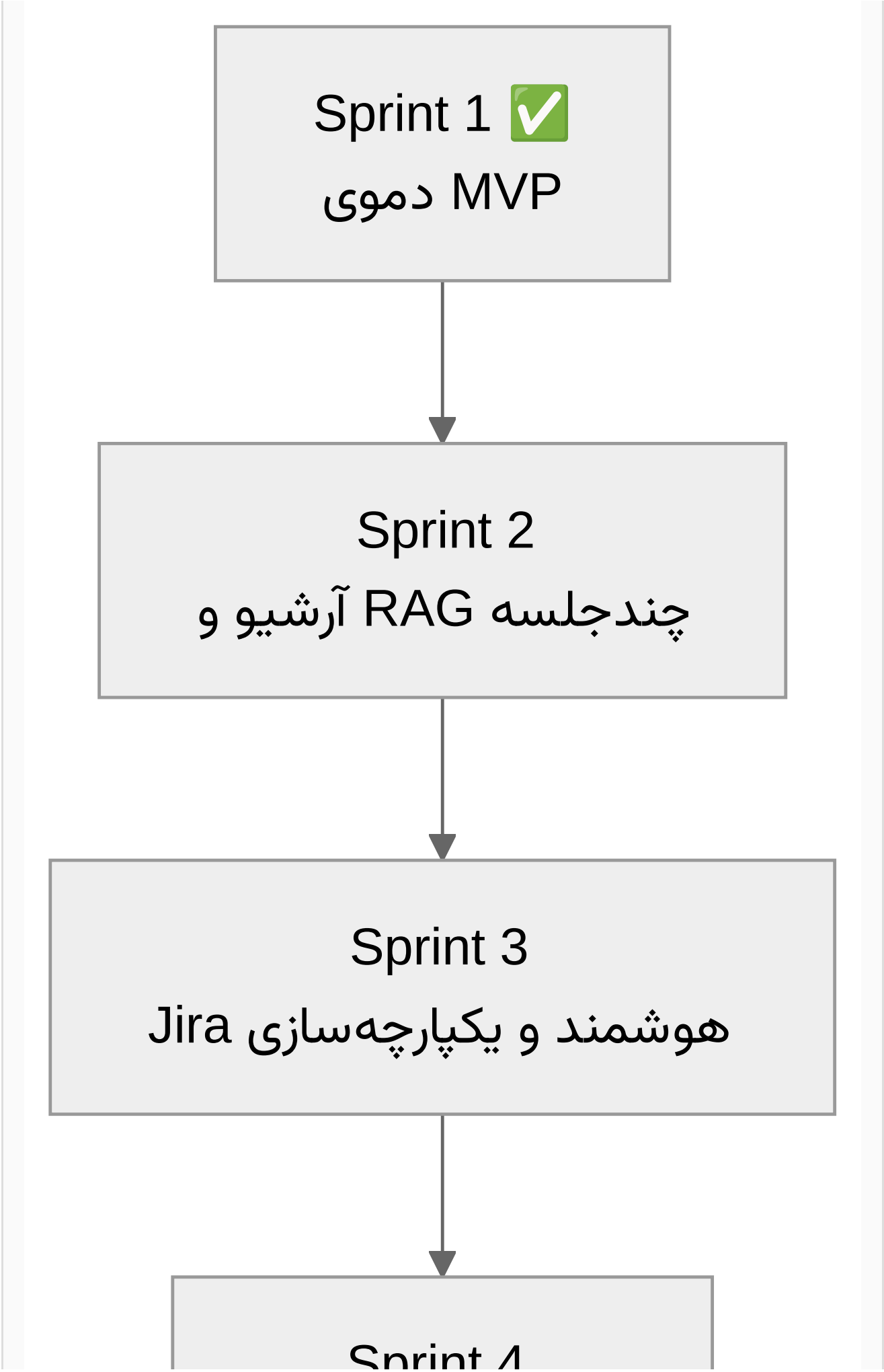
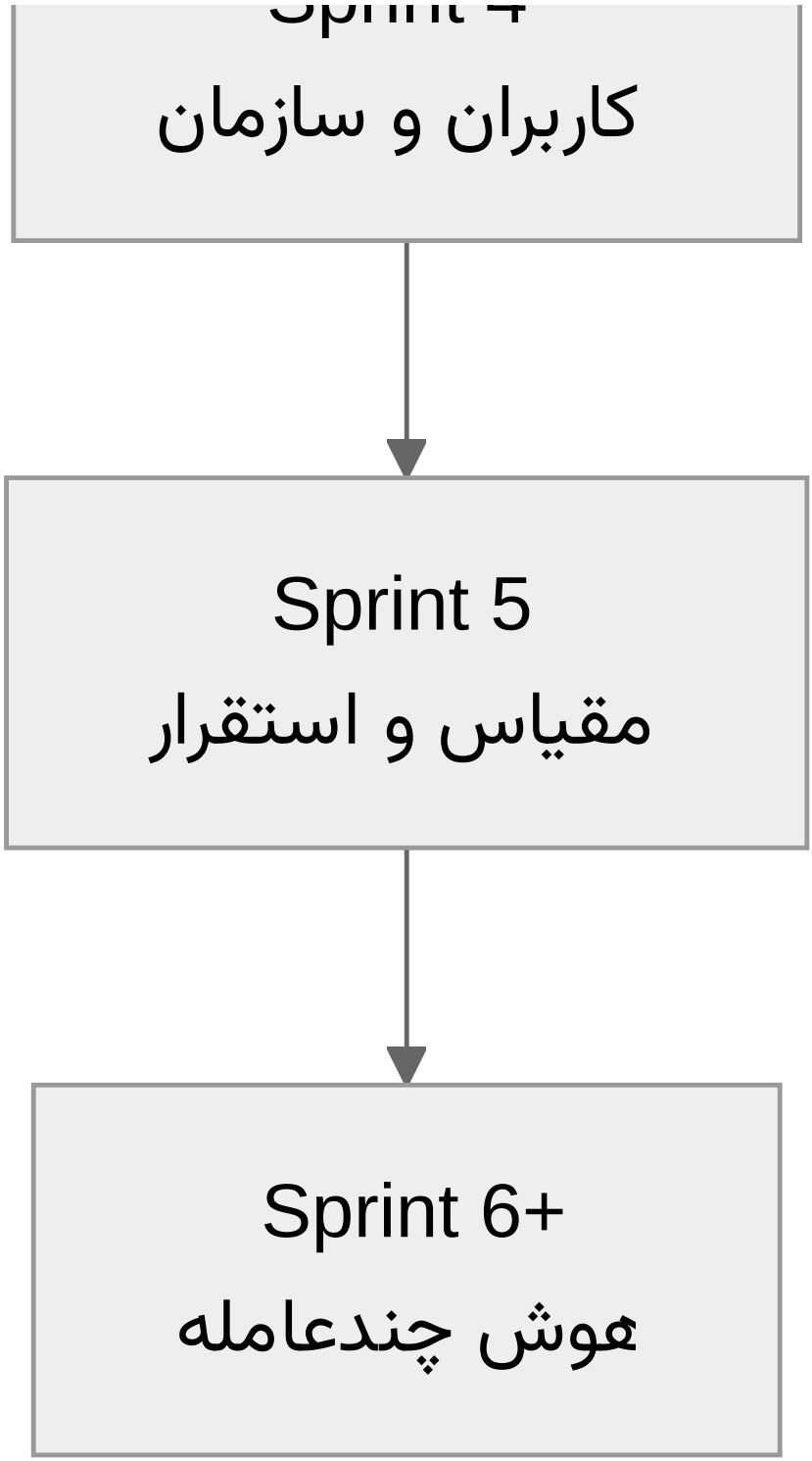
1. Roadmap at a Glance


Sprint 1 ✅
دموی MVP
Sprint 2
چندجلسه RAG آرشیو و
Sprint 3
هوشمند و یکپارچه‌سازی Jira
Sprint 4



The roadmap arc is deliberate: from one meeting (MVP) to organizational meeting memory (Sprint 2), to real operational workflow (Sprint 3), to a truly multi-tenant product (Sprints 4 and 5), and finally to differentiated multi-agent intelligence (Sprint 6+).

2. Sprint 2 — Archive & Organizational Search

Goal: Move from "one meeting" to "organizational meeting memory."

Capability	Description	Proposed implementation
Multi-meeting RAG	Query across meetings with date, project, participant filters.	Chroma metadata: `org_id`, `project`, `date`; filter in query.
Archive page	Meeting list with full-text search on title/summary.	SQLite FTS5 on `title`, `summary`.
Tags & project	Categorize meetings (standup, planning, review).	`tags`, `project_key` columns in SQLite.
Summary export	Export to Markdown / PDF for sharing with managers.	Astro template or WeasyPrint from same `MeetingAnalysis`.
Meeting comparison	"Last sprint's decisions vs now" — agent over multiple meeting_ids.	`compare_agent` + top-k from each meeting.

Acceptance criteria

- Three meetings ingested; one cross-meeting question with citations from at least two meetings.
- Archive search under 500ms (up to ~100 meetings).
- Export one meeting summary without RTL errors.

3. Sprint 3 — Jira & Operational Workflow

Goal: Tasks actually reach the right people in Jira, not anonymous issues.

Capability	Description	Technical
Assignee mapping	"Sara" in transcript ← one `accountId` in Jira.	`speaker_jira_map` table; settings UI (ready).
`jira_agent`	Preview, deduplication, epic/link suggestions.	Pydantic AI tools: `search_existing_issues`,

		`preview_create`.
Jira OAuth	Replace personal token — team-friendly.	Atlassian OAuth 2.0; encrypted refresh token storage.
Edit before submit	User edits tasks in UI, then bulk create.	Temporary state; POST only after confirmation.
Sync status	Show issue key and Jira link on meeting page.	`jira_keys_json` per task.
Issue template	Custom project fields (story points, component).	Map from `JIRA_PROJECT_KEY` + YAML config.

4. Sprint 4 — Users, Workspaces & Calendar

Goal: Real multi-user product and connection to everyday tools.

Capability	Description	Technical
Authentication	Email/password or org SSO login.	JWT session; or OIDC (Keycloak / Google Workspace).
Workspace	Each org's data isolated — meetings, speaker map, Jira.	`org_id` on all tables; tenancy middleware.
Roles (RBAC)	admin · editor · viewer — who can Jira create.	`memberships` table.
Calendar (metadata)	Import title/time/attendees from Google Calendar — transcript still paste/upload.	Calendar API read-only; pre-fill meeting form.
Summary email	Auto-send summary + meeting link to participants.	SMTP or Resend/Cloudflare Email; RTL HTML template.

Slack/Teams alerts	Short message: N new tasks + link.	Incoming webhook.

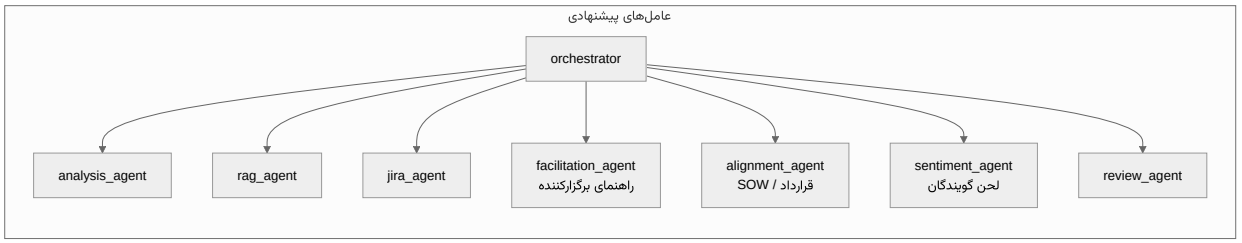
5. Sprint 5 — Scale, Quality & Deployment

Goal: Ready for organizational demo and production monitoring.

Capability	Description	Technical
PostgreSQL	Replace SQLite for multi-worker and backup.	SQLAlchemy + Alembic migration from current schema.
Vector at scale	Thousands of chunks without slowdown.	pgvector or Chroma Cloud; retention policy.
Ingest queue	Long meeting analysis in background.	Redis + worker (Celery / ARQ); status polling in UI.
Observability	Track Gemini latency and Jira errors.	OpenTelemetry + structured logs; one dashboard.
Deployment	Docker Compose; optional Cloud Run / VPS.	frontend + API + Postgres in one compose.
Eval & prompts	Golden transcript set; summary quality regression.	pytest + LLM-judge score or human rubric.
Rate limiting	Prevent API abuse.	Rate limit per org; embedding cache.

6. Sprint 6+ — Organizational Intelligence Product

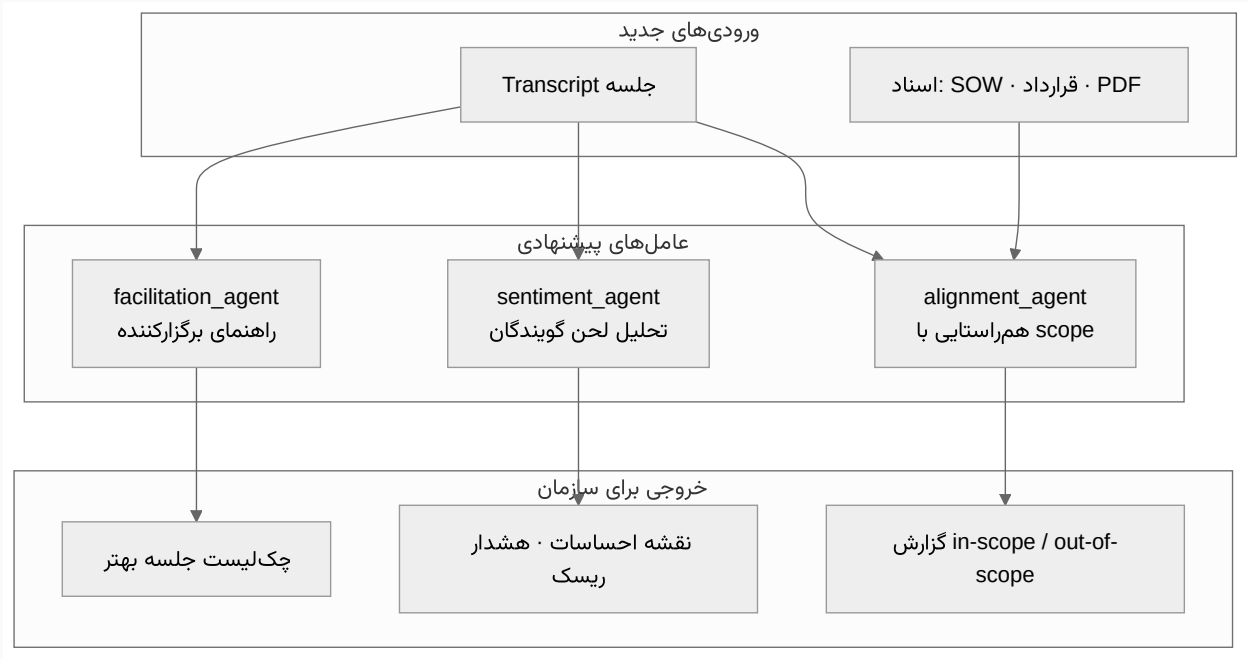
Goal: Competitive differentiation — beyond "summary + Jira."



Idea	Value
Decision tracking	Follow decisions across meetings — "Was decision X executed?"
Action items dashboard	All open tasks from all meetings, filter by assignee.
Meeting templates	Prompts tailored to standup / retro / planning.
Participation analysis	Talk ratio per speaker; one-sided meeting alerts.
Decision conflict	Agent comparing new meeting against prior decisions.
Follow-up suggestions	Auto-draft next agenda from open tasks.
Confluence / Notion	Publish minutes to org wiki.
Analysis versioning	Human edits on summary; re-run only part of pipeline.
Privacy	PII masking · retention · GDPR export.
On-prem	In-network deployment; optional self-hosted model.

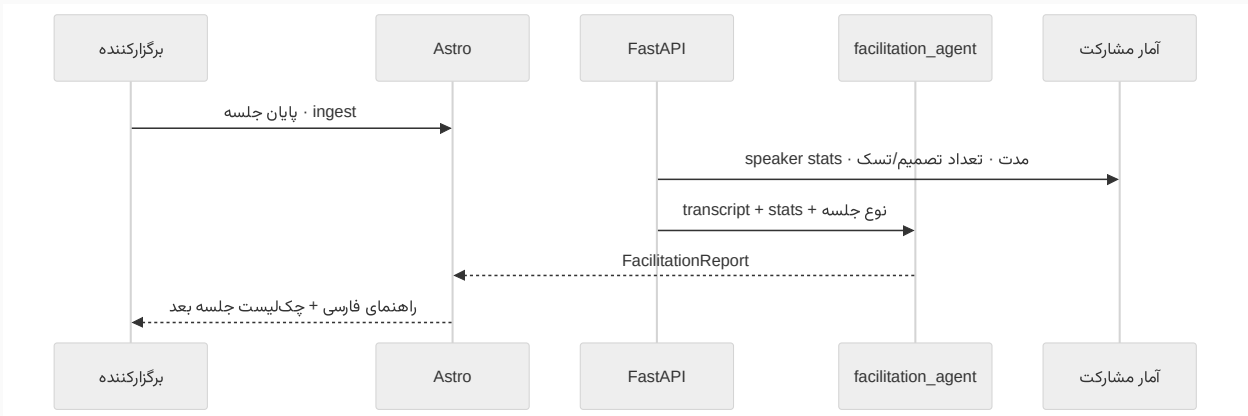
7. Advanced Capabilities (Differentiators)

Three capabilities turn the assistant from a note-taker into a meeting intelligence product. Each adds a new agent and, where needed, a new input.



7.1 Facilitator Guide *(implemented)*

After each meeting analysis, the system gives the **facilitator** practical, coaching-style suggestions: what went well, where time was wasted, and how to run the next session shorter and more effectively.



Manual checklist (today — without AI)

> **Before the meeting**
>
> ☐ Define a one-line goal and expected output (decision / task / information).

- ☐ Send agenda with approximate time per item (timebox).
- ☐ Invite only essential people — each person's role should be clear.
- ☐ Have reference documents (SOW, contract) available in advance.

During the meeting

- ☐ State decisions with "We decided that..." and name the owner.
- ☐ Park off-topic discussions in a "parking lot."
- ☐ Every 15–20 minutes, give a 30-second progress recap.

After the meeting

- ☐ Publish summary + tasks within 24 hours.
- ☐ Schedule follow-up only if open tasks remain.

FacilitationReport output

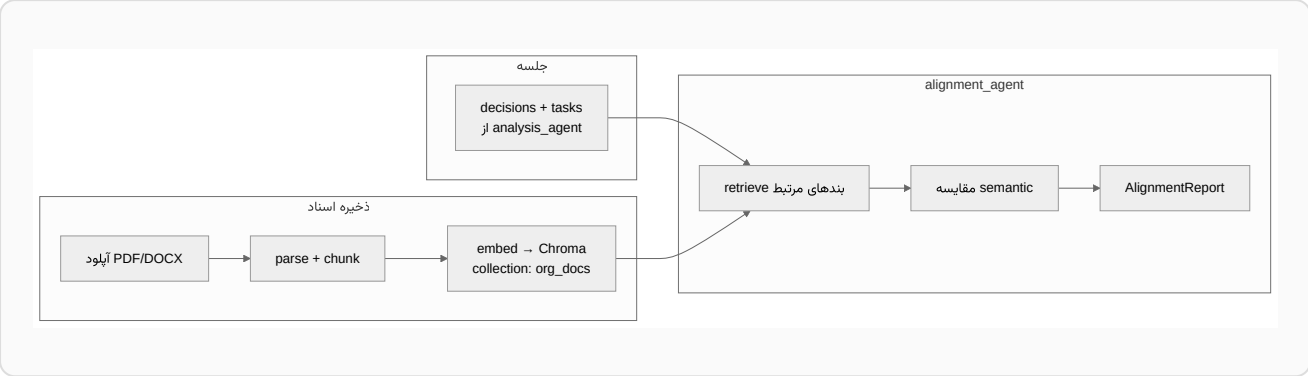
Field	Description
`what_went_well`	2–3 positives (e.g. clear decisions, balanced participation).
`improvements`	Specific suggestion (e.g. "Section X took 40% of time without a decision").
`next_meeting_agenda`	Draft agenda for next meeting from open tasks.
`timebox_suggestion`	Shorter duration or different format (e.g. async update).
`coaching_summary`	Short, encouraging summary for the facilitator.
`facilitator_score`	Optional 1–5 score based on rubric.

API: `GET /api/meetings/{id}/facilitation` — implemented, with "Meeting guide" tab in UI.

7.2 SOW / Contract Alignment *(planned)*

The organization can upload a **Statement of Work**, contract, or RFP. After each meeting, `alignment_agent` compares meeting decisions and commitments against

document clauses and labels each item **in-scope**, **out-of-scope**, or **needs legal/manager approval**.



Proposed data model

Entity	Fields
`OrgDocument`	`id`, `title`, `type` (sow|contract|rfp), `project_key`, `uploaded_at`
`AlignmentFinding`	`meeting_decision`, `doc_excerpt`, `status` (aligned|out_of_scope|unclear), `confidence`, `doc_id`
`AlignmentReport`	List of findings + Persian summary for project manager.

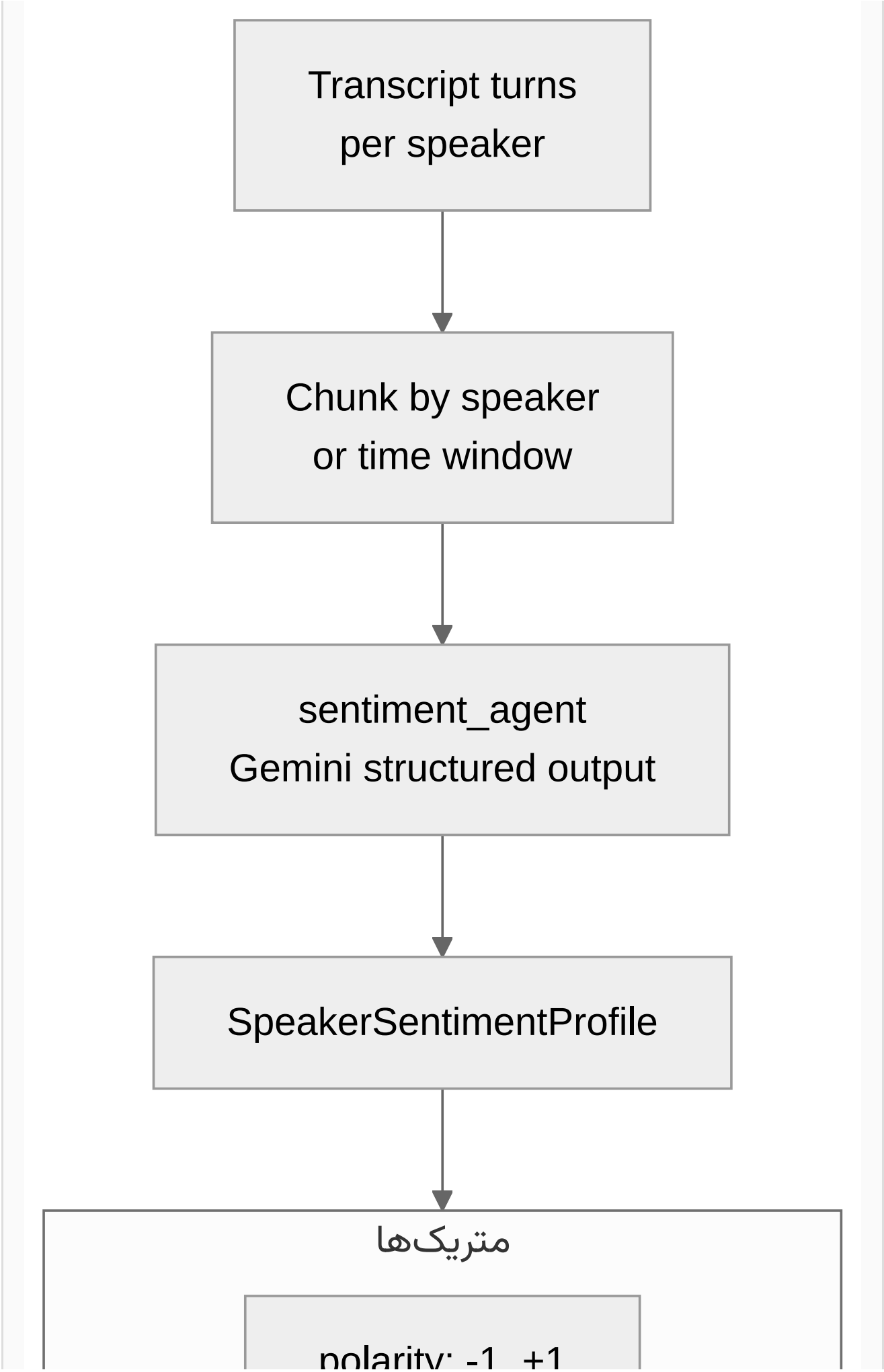
Proposed API

Method	Path	Action
POST	/api/documents	Upload SOW/contract · index in Chroma.
GET	/api/documents	List project documents.
POST	/api/meetings/{id}/alignment	Run `alignment_agent` · comparison report.

> **Product warning:** AI output is not a substitute for legal counsel — for binding contracts, human review is mandatory. The UI must show a clear disclaimer.

7.3 Sentiment Analysis *(planned)*

On **transcript text** (not raw audio), per speaker and per meeting segment: overall tone (positive / neutral / negative), **misalignment** (e.g. positive words with hidden concern), and emotional trend through the meeting.


Transcript turns
per speaker
Chunk by speaker
or time window
sentiment_agent
Gemini structured output
SpeakerSentimentProfile
متریک‌ها
polarity: -1 +1

SpeakerSentimentProfile output

Field	Meaning
`speaker`	Speaker name from transcript.
`overall_tone`	positive | neutral | negative | mixed.
`polarity_score`	Numeric from −1 to +1.
`concern_phrases`	Short quotes suggesting concern or opposition.

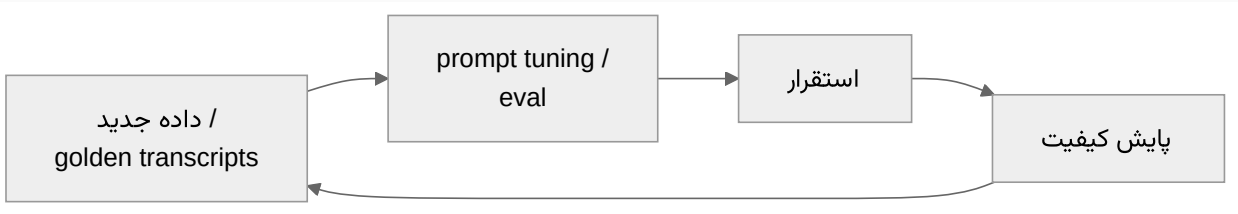
`hidden_dissatisfaction`	bool + explanation (e.g. "apparent agreement but emphasis on risk").
`timeline`	Tone shift at start / middle / end of meeting.

Proposed API: `GET /api/meetings/{id}/sentiment` — new tab beside summary/tasks/RAG.

> **Ethics & privacy:** Sentiment analysis is sensitive for team management. Only admin/facilitator should see it; org opt-in required; no negative labeling in public reports. UI framing: "signal for follow-up conversation," not "personality judgment."

8. MLOps & Quality Cycle

The MVP does not fine-tune models — it uses a hosted LLM. Therefore the operational concern is not model training but **ensuring prompt and output quality over time** — exactly where an AI product quietly degrades without fanfare.



- **Golden set:** A curated set of transcripts with reference summaries and expected tasks.
- **Automated eval:** Score each new prompt/model version against the golden set with LLM-judge and/or human rubric, as part of CI.
- **Prompt versioning:** Treat prompts like versioned artifacts so a regression can be traced to a specific change.
- **Drift monitoring:** Track summary quality, RAG citation accuracy, and Jira conversion rate over time; alert on decline.

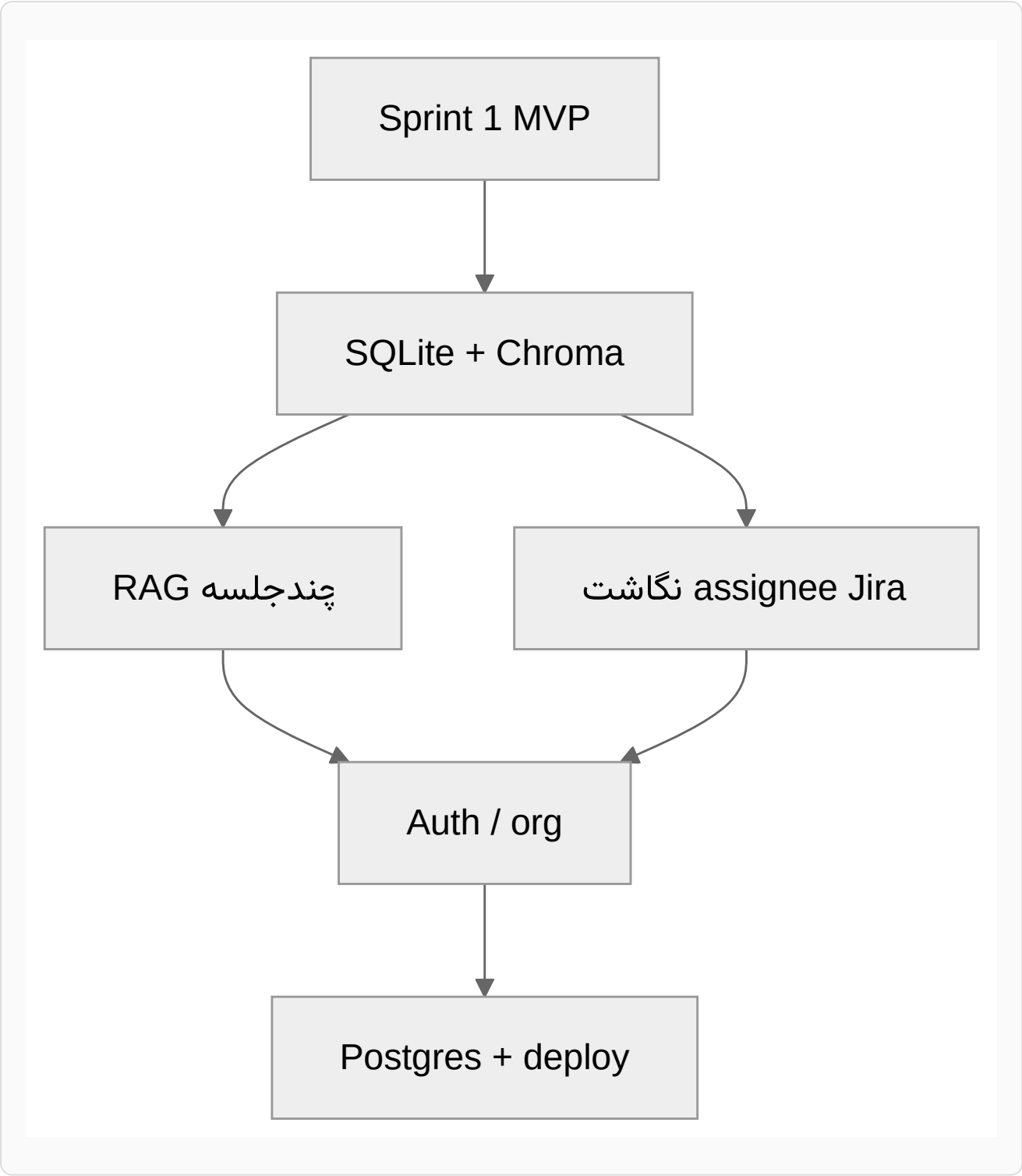
- **Logfire (initial implementation):** With Pydantic Logfire, agent runs, FastAPI/HTTPX requests, and pipeline spans (ingest → analyze → index) are traceable. Enabled with `LOGFIRE_TOKEN` in `.env`; without a token, overhead is negligible.

This work is scheduled with Sprint 5 (scale, quality, deployment) and is an honest counterpart to AIOps/MLOps theory: the lifecycle a product adopts as it matures, described realistically without overpromising in the MVP.

9. Prioritization

Priority	Item	Rationale
P0	Multi-meeting RAG + FTS archive	Highest value over MVP.
P0	Jira assignee mapping	Real pain for operational teams.
P1	Auth + workspace	Prerequisite for enterprise customer.
P1	Edit before Jira create	User trust in AI output.
P1	Facilitator guide (done) + deeper coaching	Immediate differentiation with light infra.
P1	SOW/contract alignment	Value for PMs and fixed-scope contracts.
P2	PostgreSQL + queue	When traffic exceeds demo scale.
P2	Calendar metadata + summary email	Reduce data-entry friction.
P2	Sentiment analysis	Only after RBAC and ethical opt-in.
P3	Multi-agent · Confluence	Long-term differentiation.

10. Technical Dependencies



The dependency graph explains sprint order: the storage foundation enables both multi-meeting RAG and assignee mapping; both motivate multi-tenancy (auth/org); and real tenant load eventually justifies PostgreSQL and production deployment.

11. Longer-Horizon Research Ideas

Beyond planned product work, several directions could anchor thesis research or further development:

- **Cross-meeting decision graph.** Model decisions as nodes and detect contradiction, reversal, and unfulfilled commitments across an organization's meeting history.
- **Multi-agent debate for summaries.** Use a critic/reviewer agent to challenge the first-pass summary, and measure whether adversarial review improves real faithfulness.
- **Semantic and speaker-aware chunking.** Compare turn-based vs semantic, speaker-aware chunking for retrieval accuracy on Persian transcripts.
- **Faithfulness evaluation for Persian RAG.** Build an evaluation harness for citation accuracy and hallucination rate, specific to low-resource meeting data.
- **Privacy-preserving analysis.** On-device or PII-masked pipelines so sensitive meetings can be analyzed without leaving the org network.
- **Human-in-the-loop quality.** Record human edits to summaries and tasks as feedback signals for prompt improvement and per-organization adaptation.

Related: Project Overview & Vision · Technical Implementation.